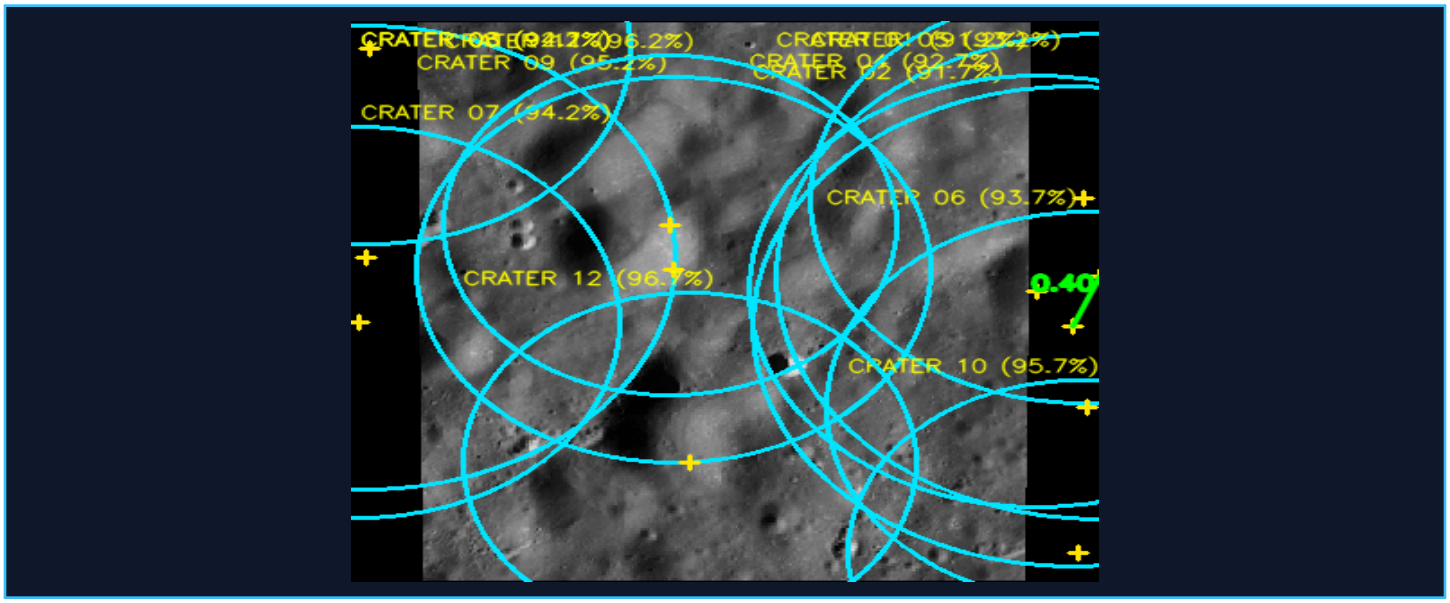
# ASTROSIGHT | PLANETARY SURFACE INTELLIGENCE

AUTONOMOUS LUNAR & MARTIAN TOPOGRAPHICAL CRATER ANALYSIS REPORT

| | | | |
|---|---|---|---|
| **Analyst / Researcher:** | Dr. Alex Reynolds | **Target Body:** | **MOON** |
| **Timestamp:** | 889d7fff-8ab5-401d- | **Resolution (GSD):** | 10.0 m/px |
| **Validation Status:** | **VALID_CRATER_DETECTED** | **Craters Identified:** | 12 |
| **Model Pipeline:** | AstroSight-CraterNet-v2.4 | **Processing Time:** | 400 ms |

## 1. TOPOGRAPHICAL DETECTION VISUALIZATION




## 2. CRATER FEATURE COORDINATES (X, Y)

| ID | Feature Name | X (px) | Y (px) | Radius (px) | Diameter (km) | Confidence |
|---|---|---|---|---|---|---|
| crater_01 | CRATER 01 | 416.0 | 189.0 | 180.0 | 3.6 km | 91.2% |
| crater_02 | CRATER 02 | 401.0 | 226.0 | 178.0 | 3.56 km | 91.7% |
| crater_03 | CRATER 03 | 8.0 | 175.0 | 172.0 | 3.44 km | 92.2% |
| crater_04 | CRATER 04 | 381.0 | 200.0 | 160.0 | 3.2 km | 92.7% |
| crater_05 | CRATER 05 | 407.0 | 131.0 | 152.0 | 3.04 km | 93.2% |
| crater_06 | CRATER 06 | 409.0 | 286.0 | 145.0 | 2.9 km | 93.7% |
| crater_07 | CRATER 07 | 4.0 | 223.0 | 145.0 | 2.9 km | 94.2% |
| crater_08 | CRATER 08 | 10.0 | 20.0 | 145.0 | 2.9 km | 94.7% |
| crater_09 | CRATER 09 | 179.0 | 184.0 | 143.0 | 2.86 km | 95.2% |
| crater_10 | CRATER 10 | 404.0 | 394.0 | 128.0 | 2.56 km | 95.7% |
| crater_11 | CRATER 11 | 177.0 | 151.0 | 126.0 | 2.52 km | 96.2% |
| crater_12 | CRATER 12 | 188.0 | 327.0 | 126.0 | 2.52 km | 96.7% |

## 3. EUCLIDEAN & REAL-WORLD SPATIAL DISTANCE MATRIX

| Vector | Origin Crater | Target Crater | Pixel Dist (px) | Real Distance (m) | Real Distance (km) |
|---|---|---|---|---|---|
| VEC-01 | CRATER 01 | CRATER 02 | 39.92 px | 399.2 m | 0.40 km |
| VEC-02 | CRATER 01 | CRATER 03 | 408.24 px | 4082.4 m | 4.08 km |
| VEC-03 | CRATER 01 | CRATER 04 | 36.69 px | 366.9 m | 0.37 km |
| VEC-04 | CRATER 02 | CRATER 03 | 396.30 px | 3962.9 m | 3.96 km |
| VEC-05 | CRATER 02 | CRATER 04 | 32.80 px | 328.0 m | 0.33 km |
| VEC-06 | CRATER 02 | CRATER 05 | 95.19 px | 951.9 m | 0.95 km |
| VEC-07 | CRATER 03 | CRATER 04 | 373.84 px | 3738.4 m | 3.74 km |
| VEC-08 | CRATER 03 | CRATER 05 | 401.42 px | 4014.2 m | 4.01 km |

## 4. SCIENTIFIC SPECIFICATIONS & UN SDG 9

**Mathematical Formulation:** Planar Euclidean metric d = √((x■ - x■)² + (y■ - y■)²) scaled via orthorectified ground sampling distance 10.0 m/px. Real-world distance D = d × resolution.
**United Nations SDG 9 Alignment:** AstroSight advances UN Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by establishing open AI infrastructure for planetary remote sensing, crater topology, and deep-space mission research.

***Scientific Disclaimer:*** *AstroSight automated crater boundary coordinates and distance vectors are generated for scientific research and mission planning. Topographical scale assumes standard planar projection for small-area orbital imagery. Always cross-correlate with high-resolution LROC NAC / HiRISE elevation models.*